Samuel Shilling

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Profile

Quantitative researcher with expertise in developing Python-based data processing pipelines, modelling complex datasets using Bayesian inference, and building simulations to predict future results. A proven track record in three peer-reviewed first author publications in a world leading astrophysical research journal, and oral presentations at five international conferences. Seeking to apply innovative scientific methods to solve challenging problems in industry.

Skills

- **Programming and tools:** Python, Linux, Git, NumPy, pandas, SciPy, Matplotlib.
- **Statistical analysis:** Data cleaning and visualisation, MCMC simulations, Spearman's rank and partial correlation tests, bootstrapping, Kolmogorov–Smirnov tests, bias tests.
- **Data modelling:** Bayesian inference, MCMC and nested sampling, posterior estimation, evidence-based model comparison, uncertainty quantification, non-linear least squares, linear regression, model extrapolation.
- **Communication:** Strong writing, verbal, and listening skills from writing three peer-reviewed publications (one in preparation), delivering presentations at five research conferences (to a range of audiences), and co-supervising four MSc student research projects.
- **Teamwork and organisation:** Experience working in teams, and leading teams, on NASA and European Space Agency projects/proposals under short deadlines, managing individual projects and PhD progress.

Relevant qualifications and experience

PhD in Astrophysics **Oct 2023 – Apr 2027 (expected)**
Lancaster University, UK

- Discovered an empirical correlation in a new data domain, proved it was not due to sample or selection biases, and quantified the properties of the correlation. Led to a publication.
- Used statistical inference to model a large sample of complex astrophysical sources for model comparison and parameter estimation. Publication in preparation.
- Presented my research at five conferences (including three major national conferences) to a range of audiences, including experts and non-experts alike.
- Co-supervised MSc Astrophysics student research projects, assisted with teaching undergraduate physics courses, and chaired the Lancaster University Astrophysics journal club in rotation.
- Worked in multi-disciplinary teams on research projects in adjacent fields to develop new approaches to current problems in the field.

Research Assistant in Astrophysics **June 2024 – Dec 2024**
NASA Goddard Space Flight Center, MD, USA

- Discovered data sampling quality was a a main driver of uncertainty in previous astrophysics works, fundamentally improving the results, and developed a simulation to predict the results in future datasets. Led to a publication.
- Analysed transient astrophysical sources, newly detected by NASA's *Swift* satellite, in real time (within minutes to hours) leading to the production of 37 community bulletin reports

MPhys Physics and Astrophysics, First Class **Sep 2018 – Jun 2023**
University of Sheffield, UK

- Strong exam based academic performance, achieving a first class degree with a final fourth year average of 78%, and a dissertation mark of 80%.
- Volunteered as the student social secretary for the Physics Society, organising social events for students.

First-author peer-reviewed publications

- Shilling et al. (in prep.) *Inferring GRB jet structures, physical parameters, and geometric properties from a large sample of X-ray afterglows.* In preparation (est. 2027).
- Shilling et al. (2026) *The intrinsic luminosity–decay correlation in subsamples of GRB X-ray afterglows.* MNRAS, 552, stag1589
- Shilling et al. (2025) *Evidence for an intrinsic luminosity-decay correlation in GRB radio afterglows.* MNRAS, 542, 2421